University of New Hampshire
Peter T. Paul College of Business and Economics

ECON 565: PREDICTIVE MODELING: DATA DRIVEN ECONOMIC ANALYSIS

SPRING, 2026

Course Information:

Course ID:	ECON 565	**Credit Hours:**	4.00
Department:		Economics	
Attributes:		Undergraduate level course	
Start/End Dates:		1/20/26 to 5/4/26	
Class Times/Location:		MW 9:40 am - 11:00 am @ Paul235	
Recommended Readings: (Optional)		Introduction to Data Analysis and Econometrics by Mark V. Siegler	

Instructor Information:

Instructor:	Bingjin (Ben) Xue Assistant Professor of Economics
Office Location/Phone:	Paul College 321 (603) 862- 3364
Office Hours:	MW 1:00 pm-2:00pm or by appointment
Email Address **Preferred Method of Contact:**	bingjin.xue@unh.edu Discussion forum on MyCourses are preferred contact for course-related questions and I will respond within one business day.

PREDICTIVE MODELING: DATA DRIVEN ECONOMIC ANALYSIS

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Course Overview

This course expands upon core topics in statistics through the study and practice of data management, data analysis, and statistical programming. Statistical programming and analytical skills are the key components of predictive modeling. Students will develop tools for collecting, organizing, interpreting, presenting, and analyzing business information. As an economics course an emphasis will be placed on how to use data to improve the information needed to make sound economic and business decisions based on marginal analysis.

Prerequisites

(ADMN 420 with a minimum grade of C- or ADMN 510 with a minimum grade of C-) and (ECON 402 with a minimum grade of C- or ECON 402H with a minimum grade of C- or EREC 411 with a minimum grade of C- or NR 411 with a minimum grade of C-).

Learning Objectives

The purpose of this course is to provide students, primarily those in business, with a conceptual introduction to analysis and its many applications. By the end of the semester, students should be able to do the following:

1. Compute and interpret basic descriptive statistical measures in Excel and Python/R
2. Understand the basic concepts of probability and utilize elementary probability rules
3. Apply techniques of statistical inference (estimation and hypothesis testing) in Excel and Python/R
4. Work with measures of statistical association (correlation and regression)
5. Utilize Excel and Python/R for data wrangling and data visualization

Textbook & Software:

Recommended Textbooks (we have *free* access, use code: **ACL-JEN-KMX**)

Introduction to Data Analysis and Econometrics by Mark V. Siegler

Note: The course primarily follows a set of lecture notes created by the professor. While textbooks are not required, they can serve as useful references for certain topics. The professor will highlight the relevant chapters in the textbooks when introducing specific subjects.

Software
MS EXCEL

Learning Management Systems:

MyCourses is the learning management tool we use for this course. The site will be used to post announcements, maintain a gradebook and provide a discussion forum.

Grades & Grading Scale

The course grade consists of the following components:

7 Homework Assignments	60%
2 In-class Exams	15% * 2 = 30%
1 Final Project (take home)	10%
Total	**100%**

** Participation is assessed by random in-class quizzes

The following scale, which is the standard UNH grading scale, determines letter grades. Grades do NOT get rounded up.

Percentage	Grade	Percentage	Grade
94–100	A	77–79.99	C+
90–93.99	A-	74-76.99	C
87–89.99	B+	70–73.99	C-
84-86.99	B	69.99 or lower	D or F
80–83.99	B-		

Assignment Details

- **Assignments:** Every *other* week, there will be one homework assignment on MyCourses, due *one week later*.

  Option: you can skip up to one of the homework assignments, and its weight will accumulate to the other one.

- **In-Class Exams:** Two in-class *written* exams will be given in-class to assess learning progress. The exam format is multiple-choice and short-answer.

- **Final Project (take-home):** There is one comprehensive final project, in which you will use all the techniques you've learned in the class to analyze a dataset and draw insights. Evaluation criteria will be posted on Canvas.

- **Attendance:** Class attendance is mandatory. Excessive absences will affect your grade for the course, so please be very careful to avoid missing any. We will be covering a considerable amount of material during each of the class meetings. Since most of the material is cumulative, a missed class will put you very far behind in a very short time. If you must miss class for any reason, If you must miss a class, please contact me beforehand.

Course Workload and Credit Hours

This syllabus reflects the federal definition of a credit hour, which entails a minimum 3 hours of engaged time per week per credit over a 15-week semester (student workload for shorter length terms must be increased proportionally per week to maintain required approximately 45 hours of work per credit per term). Examples of engaged time include class time, assignments, examinations, laboratories, participation in course-related experiences (attending a talk or performance, speakers and events, fieldwork, etc.), conferences, and office hours. Student work reflects intended learning outcomes and is verified through evidence of student achievement.

This undergraduate level course is 4 credits to reflect additional nongraded assignments and guided learning aimed at reinforcing fundamental concepts. These assignments will be given *ad hoc* throughout the semester.

Course Schedule (Topics subject to change)

Topics	Topic	Assignment
	First Day of Spring Semester	
1	Introduction and course roadmap	
2	Descriptive Statistics (Excel)	**HW1**
3	What is Probability? (Excel)	
4	Well-known Probability Distributions (Excel)	**HW2**
	In-Class Exam 1 (Date Depends on Progress)	
5	Sampling Distribution – the distribution of sample statistics (Excel)	
6	Hypothesis Testing (Excel)	**HW3**
7	Correlation and Regression Analysis (Excel)	
8	Data Cleaning in Excel	**HW4**
	SPRING RECESS	
	In-Class Exam 2 (Date Depends on Progress)	
9	From Excel to R/Python (project replicability)	

10	Data Analysis in R/Python	**HW5**
11	Data visualization in R/Python	
12	Integrate AI into your workflow	**HW6**
13	Integrate Version Control into your workflow	
14	Data extraction and SQL	**HW7**
	Last Day of Spring Semester	**Final Project**

The topics and the in-class exam dates are subject to change.

Student to Instructor Communication Expectations

How to Reach Me

Discussion Forum:

A Discussions tab can be found on MyCourses. Questions related to assignments or learning should first be asked here, if possible, so that the answers are visible to all other students. I will be active in the MyCourses class area daily, Monday through Friday. If you post a question for me in the discussion forum, please anticipate a response within 24 hours or sooner.

Email:

Bingjin.Xue@unh.edu

The use of email is reserved for questions of a private nature. If you would like to send me an email, please anticipate a response within 48 hours.

In-person Office hours:

MW 1:00-2:00 pm at Paul 321

Technical Requirements and Technical Support

See website listings for current recommendations and requirements related to this course - https://online.unh.edu/technical-requirements. Technical assistance related to Canvas is available at https://itsupport.unh.edu/mycourses/.

Academic Honesty and Plagiarism

Students are required to abide by the UNH Academic Honesty policy located in the Student Rights, Rules, and Responsibilities Handbook.

As your instructor, I proactively monitor academic integrity through regular use of tools like Turnitin and a diversified assessment approach. All works submitted to SafeAssign

become a part of a UNH proprietary database. This is actively used to identify future intellectual property theft. Plagiarism of any type may be grounds for receiving an "F" in an assignment or an "F" in the overall course. Plagiarism is defined as "the unattributed use of the ideas, evidence, or words of another person, or the conveying the false impression that the arguments and writing in a paper are your own." (UNH Academic Honesty Policy, 09.3) Examples of plagiarism include:

• Copying part or all of a previously published or written document, without quotation marks. This cut-and-paste practice is considered plagiarism, even if the writing is cited properly.
• Copying the sentence and/or paragraph structure of a source while substituting other words. "Patchwriting" is very common among students who struggle to understand information. Whether or not it is cited, this kind of paraphrasing is considered plagiarism, and it indicates that students have not fully understood the information.
• Submitting work that was completed for other courses, other times, by other students or the student him/herself.
• Submitting writing acquired from a "paper mill "resource.
UNH provides a comprehensive Plagiarism Tutorial for students who may be unclear about what specifically constitutes "plagiarism".

Incidents of academic dishonesty are reported to the school dean and may be grounds for further action. You can contact me at any time on this issue. Additional resources are located below:

http://libraryguides.unh.edu/unhmcitingsources
http://www.library.unh.edu/reference/citation.shtml
UNH Academic Honesty Policy, 09.3

Confidentiality and Mandatory Reporting of Sexual Violence or Harassment

The University of New Hampshire and its faculty are committed to assuring a safe and productive educational environment for all students and for the university as a whole. To this end, the university requires faculty members to report to the university's Title IX Coordinator (Bo Zaryckyj, Bo.Zaryckyj@unh.edu, 603-862-2930/1527 TTY) any incidents of sexual violence and harassment shared by students. If you wish to speak to a confidential support service provider who does not have this reporting responsibility because their discussions with clients are subject to legal privilege, you can contact the SHARPP Center for Interpersonal Violence Awareness, Prevention, and Advocacy at (603) 862-7233/TTY (800) 735-2964. For more information about what happens when you report, how the university treats your information once a report is made to the Title IX Coordinator, your rights and reporting options at UNH (including anonymous reporting

options), please visit student reporting options and resources easily accessible on their phones.
Help us improve our campus and community climate. If you have observed or experienced an incident of bias, discrimination or harassment, please report the incident by contacting the Civil Rights & Equity Office at UNH.civilrights@unh.edu, or TEL # (603) 862-2930 voice/ (603) 862-1527 TTY / 7-1-1 Relay NH, or visit the CREO website.
Anonymous reports may be submitted.

Emotional or mental health distress

Your academic success and overall mental health are very important. If, during the semester, you find you are experiencing emotional or mental health issues, please contact the University's (PACS) (3rd floor, Smith Hall; 603-862-2090/TTY: 7-1-1; https://www.unh.edu/pacs/) which provides counseling appointments and other mental health services. If urgent, students may call PACS M-F, 8 a.m.-5 p.m., and schedule an Urgent Same-Day Appointment.

University Disability Accommodations

According to the Americans with Disabilities Act (as amended, 2008), each student with a disability has the right to request services from UNH to accommodate their disability. If you are a student with a documented disability or believe you may have a disability that requires accommodations, please contact Student Accessibility Services (SAS); 227 Smith Hall, or sas.office@unh.edu.
Accommodation letters are created by SAS with the student. Please follow-up with your instructor as soon as possible to ensure timely implementation of the identified accommodations in the letter. Faculty have an obligation to respond once they receive official notice of accommodations from SAS, but are under no obligation to provide retroactive accommodations.
For more information, contact SAS: 227 Smith Hall, www.unh.edu/sas, 603.862.2607, 711 (Relay NH) or sas.office@unh.edu

Note: **This syllabus is subject to change. Students will be promptly notified of any changes.**